

Master Team Project WS2024

FindFul

Team 1

Member Name	Role
Sarah Majeed (sarah.majeed@informatik.hs-fulda.de)	Team Lead and Document Master
Samrat Bikram Shah	Backend Lead and Github Master
Annika Kristin Kaul	Frontend Lead
Kemer Jimenez	Team Member
Abdul Rahman Zuhaib	Team Member

Milestone 4

23/01/2024

Date Submitted	23/01/2024
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1. Product summary:

Product Name: FindFul

Major Functions:

1. Students and landlords can create an account by adding their email address, password, and username.
2. Students will be verified with their emails to have the domain “@informatik.hs-fulda.de” to ensure that they’re students of Fulda University of Applied Sciences before registration is successful.
3. Registered and Unregistered users can access our search section to perform searches using the address and postal code of the rental listing.
4. They can filter by the apartment type, size of the apartment, no. of rooms, amenities, and price range.
5. Landlords can create detailed advertisements of their listing by adding the property specifications like no. of rooms, size, amenities etc.
6. Landlords can add images to their listing for visual clarity.
7. Findful has an admin dashboard so any listings added by the landlord will first be approved by an admin before it is accessible to other users.
8. The admin can remove/ban users from the platform.
9. Students can view detailed information of a listing and can apply, either as an individual or as a group of students.
10. The students will have the option to choose from their existing groups on the app or create a new one by inviting other students.
11. On applying individually, the student can do real-time chat with the landlord of that listing.
12. On applying as a group, the group of users can have real-time chat with the landlord of that listing.

URL: <https://findful.us.to/>

2. Usability test plan:

2.1 Test objectives:

The tested function is the listing search on the FindFul homepage, including the search filters. The primary objective of the usability test is to evaluate the functionality, user experience and overall satisfaction with the search bar and filters. It is assessed how easily users can locate and interact with the search bar. The intuitivity of filter-application is tested. It is also tested how well users can interpret the results of the search.

The search is the most important function of FindFul, as it is the main navigation to specific listings. Additionally the search is embedded in FindFul's homepage. For a good first impression of FindFul, the homepage should work flawlessly to convince the user to stay on the website.

The findings from this usability test will directly influence future iterations of the search functionality, ensuring it aligns with user expectations and needs while maintaining its role as a critical component of the FindFul platform.

2.2 Test background and setup:

System setup:

The test will be conducted on a Mac with macOS 14.6, using Google Chrome.

Starting point:

At the beginning of the test, the user will be on the homepage of FindFul. No search or search filters are applied prior to the usability test. The user is not logged in.

Intended users:

The intended users of FindFul are guests who want to browse student housings in Fulda, students of the Fulda University of Applied Science looking for housing in Fulda, Landlords who want to distribute their listings and look at opponent listings, as well as Moderators of FindFul who want to make sure all listings follow FindFul's rules.

URL of the system to be tested:

<https://findful.us.to/>

What is to be measured:

The test focuses on evaluating user satisfaction, based on the ease of locating and using the search bar, usability and clarity of the filters and perceived relevance of the search results.

Usability Task description:**Task:**

Find out how many listings are available on the Leipziger Straße that cost less than 300€ warm rent and have a balcony.

User satisfaction: After completing the task, users will be asked to rate 3 statements using a 5-point Likert scale from 1: strongly disagree to 5: strongly agree:

Strongly disagree	Disagree	Neutral	Agree	Strongly agree
☐	☐	☐	☐	☐

Results:**Measurement of effectiveness:**

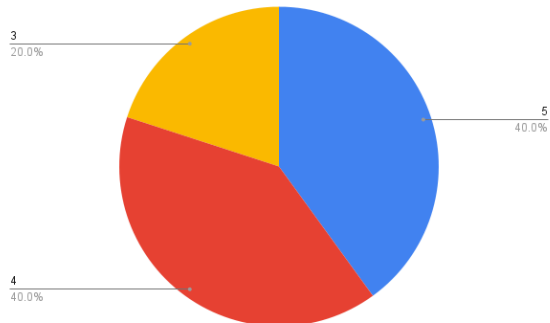
	User 1	User 2	User 3	User 4	User 5	Total
User states they found 1 listing	X	X	✓	✓	✓	3/5
The search bar and filters were used	✓	✓	✓	✓	✓	5/5
The user did not need assistance to solve the task	✓	X	✓	✓	X	3/5

Measurement of efficiency:

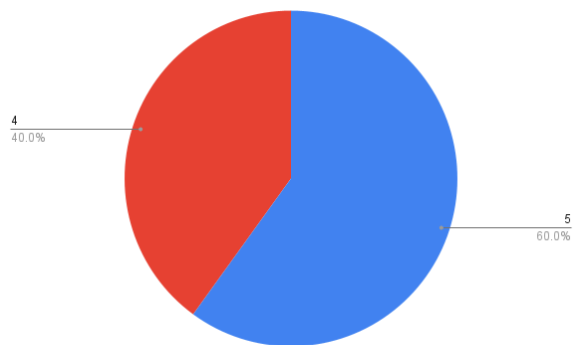
	User 1	User 2	User 3	User 4	User 5	Total
The user finishes the task in less than one minute	✓	X	X	X	X	1/5
The user needs less than 10 clicks to solve the task	✓	X	✓	✓	X	3/5

Likert scale questions:

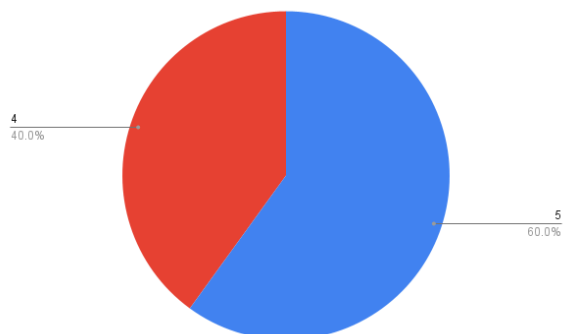
Q1: The search bar was easy to locate and use.



Q2: The use of filters was intuitive.



Q3: The design and layout of the search results page were visually appealing and helped me quickly find the information I needed.



3. QA test plan:

The objective of this test plan is to test the possible scenarios of the search functionality. These can help us conclude that the search content and their filters are working as expected. We are also checking the validation cases such as special characters not being allowed.

System setup:

The tests are conducted on a Mac with macOS 14.6, using Google Chrome. The url for the content to be tested can be found at: <https://findful.us.to/>

Test No.	Test Title	Test Description	Test Input	Expected Output	Results
1	Search Bar: By Address and Postal Code.	The search bar should search by address. For instance, if the address is Leipziger, typing “Le” in the search bar and searching should also display the result. Same logic for postal code.	“Street” or 36039	All results that have address “Street” or Postal Code 36039	Pass
2	Search Dropdown Filters: Apartment Type	The results should be filtered based on what apartment type is chosen and applied in the filter.	Single Room Apartment	All results that are single room apartments should be displayed.	Pass
3	Search Tag Filters: Amenities	Here, multiple amenities can be selected and filtered out. Based on what Amenities are selected, the results can be filtered out.	Parking and Fitted Kitchen	All results with Fitted Kitchen and Parking must be displayed.	Pass
4	Search Range	Price range of	Size: 0-40	All results	Pass

	Filters: Price, Size, Rooms	apartments, size and number of rooms available can be managed and be filtered out.	Rooms: 0-1 Price Range: 100-350	having values in the following ranges must be displayed.	
5	Search Filter: Distance from University	The distance from University is calculated based on what location the user puts in. The user can enter how many kilometers they want the accommodation to be in.	5	All results within 5 km distance of the university shall be displayed.	Pass
6	Search Box: Special Character	The Search Box should not allow special characters such as % which can potentially cause security risks.	%^	The search should not be possible and there must be a validation message displayed.	Pass

4. Code Review:

Sarah reviewed Zuhaib:

sarah-gsd reviewed 2 weeks ago View reviewed changes

frontend/findful/src/pages/profile/EditProfilePage.jsx Outdated ⚙️ Hide resolved

```
...    @@ -65,6 +79,7 @@ const EditProfilePage = () => {
65    79      await updateUserProfile(userId, updatedValues);
66    80      message.success("Profile updated successfully!");
67    81      fetchUserProfile(userId);
82    +    navigate("/");
```

sarah-gsd 2 weeks ago ...

On form submit, consider reloading the page so the user can see the changes rather than redirecting to homepage.

You can go so by replacing navigate with the following:
window.location.reload();

😊

Reply...

Unresolve conversation sarah-gsd marked this conversation as resolved.

sarah-gsd reviewed 2 weeks ago View reviewed changes

frontend/findful/src/pages/profile/EditProfilePage.jsx Outdated ⚙️ Hide resolved

```
...    @@ -51,6 +53,18 @@ const EditProfilePage = () => {
51    53      setPhoneError(value.length < 7 ? "Phone number must have at least seven digits." : "");
52    54    };
53    55
56    +    const handleAge = (_, value) => {
57    +      return new Promise((resolve, reject) => {
58    +        if (value === undefined || value === null || value === "") {
```

sarah-gsd 2 weeks ago · edited ...

This if statement is not required as we do not need to make adding age mandatory. Currently, all fields are required while editing the profile data which should not be the case.

You can change the schema as follows:

```
model Profile {
  id Int @id @default(autoincrement())
  age Int?
  gender String?
  nationality String?
  phone String? @unique
  bio String?
  user User @relation(fields: [userId], references: [id])
  userId Int @unique
}
```

Note that by adding a "?" after the attribute type, we can make the field optional.

Also note, after updating the schema, run "npx prisma migrate dev" in your backend terminal so the changes are reflected in your db.

Read -> <https://www.prisma.io/docs/orm/prisma-migrate/getting-started>

😊

Reply...

Unresolve conversation sarah-gsd marked this conversation as resolved.



sarah-gsd requested changes 2 weeks ago

[View reviewed changes](#)

sarah-gsd left a comment

...

I have added some comments for small changes.



ARZuhaib added 3 commits [last week](#)



Merge branch 'master' of <https://github.com/samratbs-fulda/gdsd> into ...

597aff0



#77: Reviewed comments for profile page

a0caef5



Merge branch 'master' of <https://github.com/samratbs-fulda/gdsd> into ...

672b47c



sarah-gsd reviewed last week

[View reviewed changes](#)

frontend/fulful/src/pages/profile/EditProfilePage.jsx Outdated

99	-	<Form.Item label="Age" name="age">
100	-	<Input type="number" />
98	+	
99	+	<Form.Item label="Age" name="age" rules={[{ validator: handleAge }]}>



sarah-gsd last week

...

The handleAge is not defined in the file as it is removed. As a result page will crash. Consider removing the following from the file:

```
rules={[{ validator: handleAge }]}
```



Reply...

Resolve conversation

Annika Reviewed Samrat:

 annikaul requested changes last month View reviewed changes

 annikaul left a comment

The searchText and card size should be reworked before merging.
Otherwise the PR looks good!



backend/src/routes/listingRouter.js Outdated

```
59 +   }
60 +   });
61 + }
62 + if (postal_code) searchText = postal_code;
```

 annikaul last month ...
The variable searchText needs to be defined beforehand. Otherwise if the postal code is empty nothing happens in line 71 (error 500).


 Reply...

Resolve conversation

frontend/findful/src/pages/Homepage.jsx Outdated

Comment on lines 177 to 179

```
177 +           style={{ width: 380,
178 +                 marginLeft: 20
179 +               }}
```

 annikaul last month ...

frontend/findful/src/pages/Homepage.jsx Outdated

Comment on lines 177 to 179

```
177 +           style={{ width: 380,
178 +                 marginLeft: 20
179 +               }}
```

 annikaul last month ...
What is the reason for this change? It is not responsive with it and the listing cards are overlapping for a smaller screen width.


 Reply...

Resolve conversation

backend/src/routes/listingRouter.js Outdated

Comment on lines 64 to 68

```
64 + if (min_price) filters.warmRent = { gt: parseFloat(min_price) };
65 + if (max_price) filters.warmRent = { ...filters.warmRent, lt: parseFloat(max_price) };
66 + if (size) filters.size = { gt: parseInt(size[0]), lt: parseInt(size[1]) };
67 + if (rooms) filters.totalRooms = { gt: parseInt(rooms[0]), lt: parseInt(rooms[1]) };
68 + if (max_distance) filters.distanceFromUni = { lt: parseFloat(max_distance) };
```

 annikaul last month ...
For some filters it might make sense to use gte (greater than or equal) or lte (lower than or equal)


 Reply...

Resolve conversation

Annika reviewed Sarah:

UI: enhance chat UI #80

Merged

annikaul merged 1 commit into master from feature/groupchat last week

Conversation 1

Commits 1

Checks 0

Files changed 2

sarah-gsd commented last week

No description provided.

UI: enhance chat UI

9df3d4c

annikaul approved these changes last week

View reviewed changes

annikaul left a comment

The new design looks great!
It might make sense though to use re-usable colors with tokens by AntDesign in the future for an easier change and corporate look of the overall design of the website.

Merged

annikaul merged commit 7af1327 into master last week

Revert

Samrat Reviewed Annika:

#116 & #115 - Feature/annikaul #81

samratbs-fulda merged 5 commits into master from feature/annikaul 20 hours ago

samratbs-fulda reviewed 2 days ago

View reviewed changes

samratbs-fulda left a comment

Overall its fine. Cleanup is recommended

frontend/findful/src/components/addListingContent/AddListingForm.jsx

256 - disabledDate={(current) => current && current.isBefore(moment(), 'day')}

257 - onChange={handleAvailableFromChange}

258 - />

311 + placeholder="Select a date"

samratbs-fulda 2 days ago

Guessing these are from formatting.

Reply...

Resolve conversation

frontend/findful/src/utis/inputValidation.js

... @@ -1,5 +1,5 @@

1 // Regex for allowed characters

2 - export const allowedCharactersRegex = /^[a-zA-Z0-9\s-.\$!,"@]*\$/;

2 + export const allowedCharactersRegex = /^[a-zA0000-9\s-.\$!,"@]*\$/;

samratbs-fulda 2 days ago

Good catch. Definitely forgot about those letters

Reply...

Resolve conversation

frontend/fulcrum/src/components/addListingContent/AddListingForm.jsx

125 + format: 'json'

126 + }).toString();

127 +

128 + const url = `https://nominatim.openstreetmap.org/search?\${query}`;

 samratbs-fulda 2 days ago • edited

Might be a good idea to delegate this call to a service or utility. Too much code in one place



 Reply...

Resolve conversation

 annika added 2 commits 20 hours ago

#114 - Add listings with real address to seed script497622d

#116 - Refactor fetching listing positionff3d4ee

 samratbs-fulda commented 20 hours ago

The changes are too valuable to not merge. Code cleanup can happen in later PRs.



Zuhaib Reviewed Kemer:

 ARZuhaib reviewed 5 minutes ago

View reviewed changes

backend/src/services/listingService.js

 ARZuhaib 5 minutes ago • edited

I see you're using very similar code in `getListingsByStatus()` (and also in other methods like `getListingsByLandlordId()`) to loop through each listing, build a thumbnail path, attempt to fetch an image and finally fall back to a default if it fails. This duplication can introduce bugs if you ever need to change the logic (for example, the fallback image or the folder path) in multiple places.

You can consider refactoring the repeated image-fetching code into a small helper function, for example:

```
async function fetchThumbnailForListing(listingId) {
  try {
    const folderKey = `${process.env.NODE_ENV}/listings/${listingId}/thumbnails`;
    return await S3Service.fetchAllImages(folderKey, { multiple: false });
  } catch (error) {
    console.error('Error fetching image for listing ${listingId}:', error);
    return await S3Service.fetchImage("image.webp");
  }
}
```

Then, inside `getListingsByStatus()` (and other methods), you can do:

```
const listingsWithImages = await Promise.all(listings.map(async (listing) => {
  const image = await fetchThumbnailForListing(listing.id);
  return { ...listing, img: image };
}));
```

This makes your code more maintainable and easier to read. If you need to change your fallback image, you only update the helper function.



 Reply...



ARZuhaib reviewed 15 minutes ago

[View reviewed changes](#)

frontend/findful/src/pages/Login/Register.jsx

 ARZuhaib 15 minutes ago

Good job on improving the password security. Just a suggestion, consider adding a small inline help tooltip or strength indicator to guide the user while typing.



 Reply...

Kemer reviewed Zuhaib:



ARZuhaib commented 2 weeks ago

[View reviewed changes](#)

No description provided.



 ARZuhaib added 6 commits 3 weeks ago

 minor bug fix 018765c

 minor bug fix cb1f236

 Merge branch 'master' of https://github.com/samratbs-fulda/gdsd into ... 72198d7

 #94: Functioning profile edit page 480f751

 merge conflicts e210737

 profileService with debugging ba156e3

 ARZuhaib requested a review from Kemer-Jimenez 2 weeks ago



 Kemer-Jimenez approved these changes 2 weeks ago

[View reviewed changes](#)

Kemer-Jimenez left a comment

Profile page get and patch working. Add redirection when changes are saved. Clean the unused profile code in the registration service.



 Kemer-Jimenez merged commit 0988937 into master 2 weeks ago

[Revert](#)

Conversation 1

Commits 2

Checks 0

Files changed 2



ARZuhaib commented on Dec 18, 2024

[View reviewed changes](#)

No description provided.



 ARZuhaib added 2 commits 2 months ago

 Add profile model and update seed script 7815f76

 Merge branch 'master' of https://github.com/samratbs-fulda/gdsd into ... 56fcb88

 ARZuhaib requested review from Kemer-Jimenez and sarah-gsd 2 months ago

 ARZuhaib closed this on Dec 18, 2024



Kemer-Jimenez requested changes on Dec 18, 2024

[View reviewed changes](#)

Kemer-Jimenez left a comment

Add profile reference in the user model.
Add userid in the profile data



5. Self-check on best practices for security:

As for the development of this full stack product, we ensure the use of the best practices for security in our three main components: Frontend, Backend and Cloud.

Frontend:

- **Dom manipulation:** Avoid direct DOM manipulation with react's virtual Dom to prevent injection attacks.
- **User inputs:** Always sanitize inputs before sending requests to the backend.
- **Https:** Always use ssl/tls for communication.

Backend:

- **RBAC (role based access control):** The access to the content is restricted depending on the user role that can be GUEST, STUDENT, LANDLORD, MODERATOR.
- **Secure Auth:** Jwt are implemented with expiration, to handle role authorization.
- **Password:** Passwords are hashed using bcrypt with salt. Passwords must be 12 characters long, contain uppercase, lowercase, numbers and special characters.
- **Environment variables:** Secrets like password salt, aws credentials and database url are stored in .env files.
- **CORS policy:** Backend service only can respond to requests from our domain.
- **ORM:** To prevent sql injections, we use prisma ORM.
- **Dependencies:** Regularly run npm audit to keep dependencies updated.

Cloud:

- **IAM roles:** Backend services communicate to other services in aws using IAM users instead to access keys to enhance security.
- **SSH:** Use secure shell protocol to connect and only with key-based authentication.
- **VCP:** Our aws database service (RDS) is not publicly accessible. It is only accessible through our private subnet.
- **Security groups:** RDS allows connections only from our ec2 instances.
- **S3 privacy:** S3 is private. Public access is denied at any instance.
- **S3 IAM:** Read and write rights are given through an IAM user.
- **Pre-signed urls:** Temporary urls are generated to give access to s3 contents.
- **Nginx for DDoS:** Nginx has an inherent protection to DDoS attacks thanks to its event-driven architecture.
- **SSL:** Ensure a secure and private communication between user and services.

6. Self-check: Adherence to original Non-functional specs – performed by team leads:

1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in Milestone 0. Application delivery shall be from chosen cloud server. **DONE**
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers. **ON TRACK**
3. All or selected application functions must render well on mobile devices. **ON TRACK**
4. Data shall be stored in the database on the team's deployment cloud server. **DONE**
5. Full resolution free media shall be downloadable directly, and full resolution media for selling shall be obtained after contacting the seller/owner. **DONE**
6. No more than 50 concurrent users shall be accessing the application at any time. **ON TRACK**
7. Privacy of users shall be protected, and all privacy policies will be appropriately communicated to the users. **ON TRACK**
8. The language used shall be English (no localization needed). **DONE**
9. Application shall be very easy to use and intuitive. **DONE**
10. Application should follow established architecture patterns. **DONE**
11. Application code and its repository shall be easy to inspect and maintain. **ON TRACK**
12. Google analytics shall be used (optional for Fulda teams). **NA**
13. No e-mail clients shall be allowed. **DONE**
14. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI. **DONE**
15. Site security: basic best practices shall be applied (as covered in the class) for main data items. **DONE**
16. Application shall be media rich (images, video etc.). Media formats shall be standard as used in the market today. **ON TRACK**
17. Modern SE processes and practices shall be used as specified in the class, including collaborative and continuous SW development. **ON TRACK**
18. For code development and management, as well as documentation like formal milestones required in the class, each team shall use their own GitHub to be set-up by class instructors and started by each team during Milestone 0. **DONE**
19. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "Fulda University of Applied Sciences Software Engineering Project, Fall 2024 For Demonstration Only" at the top of the WWW page. (Important to not confuse this with a real application). **DONE**